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Dual stage cryocooler

Abstract

A cryoablation tool has a primary fluid circuit for cryogenically cooling or heating tissue surrounding the distal portion of the cryoablation tool. The primary fluid circuit has a primary-primary heat exchanger facilitating recuperative heat exchange between a high pressure stream of a primary fluid and a low pressure stream of the primary fluid. The cryoablation tool has a secondary fluid circuit having a secondary-secondary heat exchanger for permitting recuperative heat exchange between a high pressure stream of a secondary fluid and a low pressure stream of the secondary fluid. The secondary fluid circuit also has a primary-secondary heat exchanger permitting heat exchange between the high pressure stream of the primary fluid and the low pressure stream of the secondary fluid.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims priority to and is a divisional application of U.S. Ser. No. 16/994,110, filed Aug. 14, 2020, which claims priority to U.S. Provisional Application No. 62/886,853, filed Aug. 14, 2019, the contents of which are herein incorporated by reference in their entirety.

BACKGROUND

(1) This disclosure relates to a cryoprobe for use in cryoablation and a system for cryoablation.

(2) During cryosurgery, a surgeon may deploy one or more cryoprobes to ablate a target area of a patient anatomy by freezing and thawing the tissue. In one example, a cryoprobe uses the Joule-Thomson effect to produce cooling or heating of the probe tip. In such cases, the expansion of a cryofluid in the cryoprobe from a higher pressure to a lower pressure leads to cooling of the device tip to temperatures at or below those corresponding to cryoablating a tissue in the vicinity of the tip. Heat transfer between the expanded cryofluid and the outer walls of the cryoprobe leads to formation of an ice ball, in the tissue around the tip and consequent cryoablation the tissue.

(3) Some cryoprobes may be useful for ablating lesions in a human lung or other body passageways. In such cases, the cryoprobes may have to navigate tortuous and/or narrow passageways, and accordingly may be constructed such that the cryoprobe may not support a cryofluid pressurized to pressures typical for cryoablation (e.g., 3500 psi). Further, delivering cryofluid at pressures lower than typical supply pressure for cryoablation (e.g., 3500 psi) may not result in sufficient cooling or formation of ice balls of desirable sizes within desirable times (e.g., 35 mm ice ball in 10 minutes).

SUMMARY

(4) Advantageous aspects of the present disclosure provide a cryoablation tool with a two-stage cryocooler suitable for producing ice balls of sizes suitable for cryoablation.

(5) In a first aspect of the cryoablation tool, a cryoablation tool is provided comprising: a primary fluid circuit comprising a flow of a high pressure stream of a primary fluid and a low pressure stream of the primary fluid, the primary fluid circuit being fluidly coupled to a distal portion of the cryoablation tool for cryogenically cooling or heating tissue surrounding the distal portion of the cryoablation tool, the primary fluid circuit comprising a primary-primary heat exchanger configured for recuperative heat exchange between the high pressure stream of the primary fluid and the low pressure stream of the primary fluid; a secondary fluid circuit comprising a high pressure region for flow of a high pressure stream of a secondary fluid and a low pressure region for flow of a low pressure stream of the secondary fluid, the secondary fluid circuit comprising a secondary-secondary heat exchanger for recuperative heat exchange between the high pressure stream of the secondary fluid and the low pressure stream of the secondary fluid, the secondary-secondary heat exchanger terminating in a secondary-secondary outlet; and a primary-secondary heat exchanger configured for recuperative heat exchange between the high pressure stream of the primary fluid and the low pressure stream of the secondary fluid, the primary-secondary heat exchanger comprising a primary-secondary inlet, the primary-secondary heat exchanger being arranged such that the primary-secondary inlet is positioned downstream of the secondary-secondary outlet along a direction of flow of the high pressure stream of the secondary fluid, the primary-secondary heat exchanger and the secondary-secondary heat exchanger arranged such that the low pressure stream of the secondary fluid exchanges heat with the primary fluid in the primary-secondary heat exchanger prior to exchanging heat with the high pressure stream of the secondary fluid in the secondary-secondary heat exchanger.

(6) According to advantageous aspects, each of the primary-primary heat exchanger, the secondary-secondary heat exchanger and the primary-secondary heat exchanger comprises a tube having an outer wall. The outer wall of the tube comprises extensions to increase a surface area of the outer wall.

(7) In another aspect, each of the primary-primary heat exchanger, the secondary-secondary heat exchanger and the primary-secondary heat exchanger comprises a finned tube.

(8) In a further aspect, the secondary-secondary heat exchanger and the primary-secondary heat exchanger each comprises a tube that is wound in a series of turns about a tubular mandrel.

(9) In one approach, the secondary-secondary heat exchanger and the primary-secondary heat exchanger are each wound about the same tubular mandrel. The primary-primary heat exchanger may also comprise a tube that is wound in a series of turns about a tubular mandrel.

(10) According to an aspect of this disclosure, the primary-primary heat exchanger is fluidly

coupled with the primary-secondary heat exchanger such that the high pressure stream of the primary fluid flows through the primary-secondary heat exchanger prior to flowing through the primary-primary heat exchanger.

(11) In another aspect, the fluid coupling between the primary-primary heat exchanger and the primary-secondary heat exchanger is configured to fluidly isolate the low pressure stream of the primary fluid from the primary-secondary heat exchanger.

(12) In an additional aspect, the fluid coupling between the primary-primary heat exchanger and the primary-secondary heat exchanger is configured to fluidly isolate the low pressure stream of the primary fluid from the secondary-secondary heat exchanger.

(13) In a further aspect, the flow of the low pressure stream of the primary fluid is isolated from the low pressure region of the secondary fluid circuit.

(14) In a further aspect, the low pressure stream of the primary fluid may be arranged concentrically with the low pressure flow of the secondary fluid and is physically separated therefrom.

(15) In a further aspect, the high pressure region of the primary-secondary heat exchanger and the high pressure region of the secondary-secondary heat exchanger are isolated from the low pressure stream of the primary fluid.

(16) In a further aspect, both the secondary-secondary heat exchanger and the primary-secondary heat exchanger are isolated from the low pressure region of the primary fluid circuit.

(17) According to an aspect, the primary fluid circuit comprises a first Joule-Thomson (J-T) orifice positioned in the distal section of the cryoablation tool, the first J-T orifice being fluidly coupled to the primary-primary heat exchanger via a primary supply conduit to receive the high pressure stream of the primary fluid after the high pressure stream of the primary fluid passes through the primary-primary heat exchanger, the first J-T orifice configured to cryogenically expand the high pressure stream of the primary fluid to the low pressure stream of the primary fluid.

(18) In a further aspect, the cryoablation tool further comprises a primary return conduit for passage of the low pressure stream of the primary fluid therethrough, the first J-T orifice being disposed within the primary return conduit.

(19) In certain aspects, the secondary fluid circuit comprises a second Joule-Thomson (J-T) orifice, the second J-T orifice being fluidly coupled to the secondary-secondary heat exchanger via a secondary supply conduit to receive the high pressure stream of the secondary fluid after the high pressure stream of the secondary fluid passes through the secondary-secondary heat exchanger, the second J-T orifice configured to cryogenically expand the high pressure stream of the secondary fluid to the low pressure stream of the secondary fluid.

(20) In an additional aspect, the cryoablation tool further comprises a second return conduit to permit passage of the expanded secondary fluid, wherein, the secondary-secondary heat exchanger, the primary-secondary heat exchanger, and the second J-T orifice are each disposed within the second return conduit.

(21) According to an aspect, the second return conduit is fluidly isolated from the low pressure stream of the primary fluid of the primary fluid circuit.

(22) Further, the low pressure stream of the primary fluid and/or the secondary fluid may be configured to vent to atmosphere. A common vent conduit may be configured for the low pressure stream of the primary fluid and/or the secondary fluid to vent to the atmosphere (via the common vent conduit).

(23) In an aspect, the cryoablation tool includes a primary inlet conduit for the primary fluid circuit to deliver the primary fluid to the primary fluid circuit, and a secondary inlet conduit to deliver the secondary fluid to the secondary fluid circuit. The common vent conduit may be arranged to pass the primary fluid or the secondary fluid over the primary inlet conduit and the secondary inlet conduit before venting the primary fluid or the secondary fluid to the atmosphere.

(24) A further embodiment provides A cryoablation tool comprising: a shaft having a proximal end

and a distal end; a primary supply conduit configured to supply a primary fluid from a high-pressure cryogenic gas source to a distal end of the shaft, the primary supply conduit having a first Joule-Thomson orifice at a distal end thereof, configured to deliver the primary fluid to a first expansion chamber; a primary return conduit configured to carry the primary fluid away from the distal expansion chamber; a secondary supply conduit configured to supply a secondary fluid to a second Joule-Thomson orifice, the second J-T orifice configured to deliver the secondary fluid to a second expansion chamber; a secondary return conduit configured to carry the secondary fluid away from the second expansion chamber; the primary supply conduit comprising a primary-secondary heat exchange region and a primary-primary heat exchange region, the primary-primary heat exchange region being down stream of the primary-secondary heat exchange region; and the secondary supply conduit comprising a secondary-secondary heat exchange region up stream of the second J-T orifice; the primary fluid return conduit further configured to pass the primary cryogenic gas over the primary-primary heat exchange region, in counterflow to the supply direction; the secondary return conduit further configured to pass the secondary fluid sequentially over the primary-secondary heat exchange region and the secondary-secondary heat exchange region in counterflow to the supply direction in each case.

(25) In a further aspect, the primary supply conduit is arranged in a first plurality of coils, configured as a primary-secondary heat exchanger within the primary-secondary heat exchange region and a second plurality of coils, configured as a primary-primary heat exchanger, within the primary-primary heat exchange region; and the secondary supply conduit being arranged in a third plurality of coils configured as a secondary-secondary heat exchanger within the secondary-secondary heat exchange region.

(26) In a further aspect, the primary return conduit is configured to pass the expanded primary fluid over the primary-primary heat exchanger; and the secondary return conduit is configured to pass the expanded secondary fluid over the primary-secondary heat exchanger prior to passing the expanded secondary fluid over the secondary-secondary heat exchanger.

(27) In a further aspect, the first plurality of coils comprises a first coil and a second coil, the first coil arranged upstream of the second coil relative to the direction of primary fluid flow through the first coil; each coil of the second plurality of coils being positioned downstream of the said second coil relative to the direction of primary fluid flow in the first coil; and each coil of the third plurality of coils is positioned upstream of the first coil relative to the direction of primary fluid flow in the said first coil.

(28) In a further aspect, the secondary return conduit may be configured to pass the expanded secondary fluid, sequentially over the full length of the primary-secondary heat exchange region before passing the secondary fluid over the full length of the secondary-secondary heat exchange region. The secondary return conduit may be configured to pass the expanded secondary fluid over each coil of the primary-secondary heat exchanger before passing it over the secondary to secondary heat exchanger.

(29) In a further aspect the primary-primary heat exchange region is arranged within the primary return conduit and is further, arranged distally of both the primary-secondary and the secondary-secondary heat exchange regions.

(30) In a further aspect the secondary-secondary heat exchange region is arranged proximally to the primary-secondary heat exchange region, within the secondary return conduit.

(31) In a further aspect the primary supply conduit comprises a tubular region wound in a series of turns about a mandrel to form the primary-secondary heat exchanger. The secondary supply conduit may comprise a tubular region wound in a series of turns about a mandrel to form the secondary-secondary heat exchanger. Further, the primary supply conduit may comprise a tubular region wound in a series of turns about a mandrel to form the primary-secondary heat exchanger, and the secondary supply conduit may comprise a tubular region wound in a series of turns about the same mandrel to form the secondary-secondary heat exchanger. The primary supply conduit may

comprise a tubular region wound in a series of turns about a mandrel to form the primary-primary heat exchanger.

(32) In a further aspect the primary return conduit is configured to pass the primary fluid over the primary-primary heat exchange region but not the primary-secondary heat exchange region or the secondary-secondary heat exchange region.

(33) In a further aspect the primary-secondary heat exchange region and the secondary-secondary heat exchange region are each disposed within the secondary return conduit. The secondary return conduit may then fluidly isolate the primary return conduit from the primary-secondary heat exchange region and the secondary-secondary heat exchange region.

(34) In a further aspect the primary return conduit comprises a portion disposed concentrically about the secondary return conduit. This portion may be downstream of the primary-primary heat exchange region with respect to the direction of expanded primary fluid flow.

(35) In a further aspect the primary return conduit is configured to vent the primary fluid to atmosphere and/or the secondary return conduit is configured to vent the secondary fluid to atmosphere.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an exemplary schematic of a cryoablation tool;

(2) FIG. 2 is another exemplary schematic of the cryoablation tool of FIG. 1 illustrating interior details thereof;

(3) FIG. 3 is a pictorial view of an exemplary precooler of the cryoablation tool of FIG. 1;

(4) FIG. 4 is a pictorial view of an exemplary primary-primary heat exchanger of the cryoablation tool of FIG. 1;

(5) FIG. 5 is a thermodynamic schematic illustrating the primary-primary, primary-secondary and secondary-secondary heat exchangers associated with the cryoablation tool of FIG. 1; and

(6) FIG. 6 is an exemplary thermodynamic property diagram illustrating the thermodynamic states associated with the primary-primary, primary-secondary and secondary-secondary heat exchangers illustrated in FIG. 1.

DETAILED DESCRIPTION

(7) FIG. 1 illustrates a schematic of a cryoablation tool **100** according to an embodiment. According to some embodiments, the cryoablation tool **100** may include a catheter **102**. In advantageous aspects, the catheter **102** may be inserted into a working channel of a bronchoscope, and may, therefore be, generally flexible. The cryoablation tool **100** may include a distal section **104** and a proximal section **106**. The distal section **104** may terminate in a distal operating tip **108**. In some instances, the distal operating tip **108** may have to pierce tissue, and therefore, may be configured as a rigid tip. Alternatively, the distal operating tip **108** may not be a rigid tip.

(8) With reference to FIGS. 1 and 2, the cryoablation tool **100** includes a primary fluid circuit **110** (shown in solid lines) **110**. The primary fluid circuit **110** may be associated with a primary fluid. The primary fluid circuit **110** may be in fluid communication with the distal section **104** of the cryoablation tool **100**. The primary fluid may cool (e.g., cryogenically) or heat tissue surrounding the distal portion of the cryoablation tool **100**. The primary fluid circuit **110** may include a primary supply conduit **112** for carrying a high pressure stream of the primary fluid. The primary fluid circuit **110** may also include a primary return conduit **114** for carrying a low pressure stream of the primary fluid.

(9) The primary fluid circuit **110** may also include a primary-primary heat exchanger **120** configured for heat exchange (e.g., recuperative heat exchange) between the high pressure stream of the primary fluid and the low pressure stream of the primary fluid. The primary-primary heat

exchanger **120** includes a primary-primary inlet **122** and a primary-primary outlet **124**. The primary-primary inlet **122** may be fluidly coupled to the primary supply conduit **112** to receive the high pressure stream of the primary fluid. The primary-primary outlet **124** may be fluidly coupled to a primary conduit **128** (e.g., a primary capillary tube) to deliver the primary fluid toward the distal operating tip **108**.

(10) In advantageous aspects, the high pressure stream of the primary fluid (from the primary supply conduit **112**) may flow through a first cryocooler **130**. In such embodiments, the high pressure stream of the primary fluid may be upstream (with respect to direction **166**) of the first cryocooler **130**. Further, the low pressure stream of the primary fluid may be downstream (with respect to direction **166**) of the first cryocooler **130**. As illustrated in FIG. 2, the primary supply conduit **128** (e.g., primary capillary tube) comprises a first Joule-Thomson (“J-T”) orifice **130** downstream of the primary-primary outlet, at the end of the primary supply conduit. Accordingly, the first cryocooler **130** can be an open loop cryocooler, such as the first J-T orifice **130**.

(11) In such embodiments, the high pressure stream of the primary fluid may undergo expansion at or downstream of the first J-T orifice **130** in a first expansion chamber **132**. The first expansion chamber **132** may be in fluid communication with the primary return conduit **114** to carry the (expanded low pressure stream of) the primary fluid away from the expansion chamber, toward the proximal section **106** (e.g., to vent to atmosphere, if the primary fluid circuit **110** is an open circuit, or back to a primary fluid source if the primary fluid circuit **110** is a closed circuit).

(12) The primary fluid can be a cooling fluid (e.g., nitrogen, air, argon, krypton, xenon, N.sub.2O, CO.sub.2, CF.sub.4) in embodiments. In such cases, the high pressure stream of the primary fluid may be at a pressure such that expansion via the first J-T orifice **130** may result in the primary fluid cooling to temperatures for cryogenically ablating tissue surrounding the distal operating fluid. In certain aspects, the pressure of the high pressure stream of the primary fluid upstream of the first J-T orifice **130** can be between about 1000 psi and about 2000 psi (e.g., about 1800 psi).

Accordingly, in the embodiments where the primary fluid is a cooling fluid, the temperature of the primary fluid after expansion from the first J-T orifice **130** can be less than about 190 Kelvin.

(13) Alternatively, the primary fluid can be a heating fluid (e.g., helium, hydrogen). In such cases, the high pressure stream of the primary fluid may be at a pressure such that expansion via the first J-T orifice **130** may result in a temperature increase of the primary fluid, correspondingly resulting in heating of tissue surrounding the distal operating fluid. Such embodiments may be useful for thawing frozen tissue.

(14) In some embodiments, the primary fluid circuit **110** may be arranged such that the flow of the high pressure stream of the primary fluid is countercurrent to the flow of the low pressure stream of the primary fluid over at least certain portions of the primary fluid circuit **110**. For instance, as mentioned above, in an embodiment, the high pressure stream of the primary fluid passes through a first J-T orifice **130**. As illustrated in FIGS. 1 and 2, the flow direction of the primary fluid is reversed upon expansion at the first J-T orifice **130**, thereby resulting in countercurrent flow of the high pressure stream of the primary fluid and the low pressure stream of the primary fluid near the first J-T orifice **130**. Furthermore, the high pressure stream of the primary fluid and the low pressure stream of the primary fluid may be in countercurrent flow near the primary-primary inlet **122** of the primary-primary heat exchanger **120**, so the primary return conduit is configured to pass the primary cryogenic gas over the primary-primary heat exchange region, in counterflow to the supply direction.

(15) According to advantageous aspects, the countercurrent arrangement of the high pressure stream of the primary fluid and the low pressure stream of the primary fluid may permit recuperative heat exchange therebetween. In instances where the primary fluid circuit **110** carries a cooling gas, the recuperative heat exchange may include heat removal from the high pressure stream of the primary fluid leading to precooling of the primary fluid prior to expansion via the first J-T orifice **130**. In instances where the primary fluid circuit **110** carries a heating gas, the

recuperative heat exchange may include heat addition from the high pressure stream of the primary fluid leading to preheating of the primary fluid prior to expansion via the first J-T orifice **130**. The recuperative heat exchange may advantageously lead to reaching desired temperatures at the distal operating tip **108**.

(16) As described previously, in some embodiments, the cryoablation tool **100** may include a catheter **102** sufficiently flexible to be maneuvered within a bronchoscope. In such embodiments, the catheter **102** may be made of materials that may not be able to withstand typical pressures at which the primary fluid may be supplied in other cryoablation tools (e.g., greater than about 2000 psi, for instance, about 3500 psi). Under such circumstances, it may be advantageous to supply the primary fluid at pressures less than typical pressures for cryoablation. The primary fluid may be supplied at less than about 2000 psi, for instance, about 1800 psi). However, to facilitate temperatures suitable for cryoablation from being reached at the distal operating tip **108**, a secondary fluid circuit **140** may be provided to precool the primary fluid, as will be described further below.

(17) Referring to FIGS. **1** and **2**, the cryoablation tool **100** includes a secondary fluid circuit **140** (shown in dotted lines). The secondary fluid circuit **140** may include a secondary supply conduit **142**, which carries a high pressure stream of a secondary fluid. The secondary supply conduit is configured to supply a second fluid to a second cryocooler (J-T orifice). The second cryocooler may be configured to deliver the second fluid to a second expansion chamber. The secondary fluid circuit **140** may also include a secondary return conduit **144** which carries a low pressure stream of the secondary fluid. The secondary return conduit may be configured to carry the second fluid away from the second expansion chamber.

(18) In advantageous aspects, the secondary fluid circuit **140** may facilitate heat exchange between the primary fluid and the secondary fluid. In some such advantageous aspects, the secondary fluid circuit **140** can be used to precool the high pressure stream of the primary fluid in embodiments where the primary fluid cools upon expansion to cryogenically ablate tissue surrounding the distal operating tip **108**. In such advantageous aspects, referencing FIG. **2**, the secondary fluid circuit **140** may also include a primary-secondary heat exchanger **150**. In certain aspects, the primary-secondary heat exchanger **150** may facilitate heat exchange between the high pressure stream of the primary fluid and the low pressure stream of the secondary fluid. The primary-secondary heat exchanger **150** may include a primary-secondary inlet **152**, and a primary-secondary outlet **154**.

(19) With continued reference to FIG. **2**, the secondary fluid circuit **140** may include a secondary-secondary heat exchanger **160** that permits heat exchange between the high pressure stream of the secondary fluid and the low pressure stream of the secondary fluid (e.g., recuperative heat exchange). In some advantageous aspects, the secondary fluid may also be a cooling fluid. In such embodiments, recuperative heat exchange between the high pressure stream of the secondary fluid and the low pressure stream of the secondary fluid may remove heat from the high pressure stream of the secondary fluid. Accordingly, the secondary-secondary heat exchanger **160** may facilitate precooling the high pressure stream of the secondary fluid.

(20) Referring again to FIG. **2**, the secondary-secondary heat exchanger **160** may include a secondary-secondary inlet **162** and a secondary-secondary outlet **164**. The secondary-secondary inlet **162** may be upstream of the secondary-secondary outlet **164**, along a direction **166**. Further, the secondary-secondary inlet **162** may be upstream of each of the primary-primary inlet **122**, and the primary-primary outlet **124**. Additionally, the secondary-secondary outlet **164** may also be upstream of each of the primary-primary inlet **122**, and the primary-primary outlet **124**.

(21) The secondary-secondary inlet **162** may be fluidly coupled to the secondary supply conduit **142** to receive the high pressure stream of the secondary fluid. The secondary-secondary outlet **164** may be fluidly coupled to a secondary conduit **168** (e.g., a secondary capillary tube **168**) to receive the secondary fluid from the secondary-secondary outlet **164**. The secondary fluid may thus pass through the secondary-secondary heat exchanger **160** by entering at the secondary-secondary inlet

162, and exiting at the secondary-secondary outlet **164**.

(22) In advantageous aspects, the high pressure stream of the secondary fluid leaving the secondary-secondary outlet **164** may enter the secondary capillary tube **168**, and may flow through a second cryocooler **170**. In such embodiments, the high pressure stream of the secondary fluid may be upstream (with respect to direction **166**) of the second cryocooler **170**. Further, the low pressure stream of the secondary fluid may be downstream (with respect to direction **166**) of the second cryocooler **170**. In the illustrated embodiment of FIG. 2, the secondary conduit **168** (e.g., secondary capillary tube **168**) downstream of the secondary-secondary outlet **164** terminates in a second Joule-Thomson (“J-T”) orifice **170**. Accordingly, the second cryocooler **170** can be the second J-T orifice **170**. In such embodiments, the high pressure stream of the secondary fluid may undergo expansion at or downstream of the second J-T orifice **170** in a second expansion chamber **172**. The second expansion chamber **172** may be in fluid communication with the secondary return conduit **144** to carry the expanded low pressure stream of the secondary fluid (e.g., to vent to atmosphere, if the secondary fluid circuit **140** is an open circuit, or back to a secondary fluid source if the secondary fluid circuit **140** is a closed circuit).

(23) In some embodiments, the secondary fluid circuit **140** may be arranged such that the flow of high pressure stream of the secondary fluid is countercurrent to the flow of the low pressure stream of the secondary fluid over at least certain portions of the secondary fluid circuit **140**. For instance, as mentioned above, in an embodiment, the high pressure stream of the secondary fluid passes through a second J-T orifice **170**. As illustrated in FIGS. 1 and 2, the flow direction of the secondary fluid is reversed upon expansion at the second J-T orifice **170**, thereby resulting in countercurrent flow of the high pressure stream of the secondary fluid and the low pressure stream of the secondary fluid near the second J-T orifice **170**. Furthermore, the high pressure stream of the secondary fluid and the low pressure stream of the secondary fluid may be in countercurrent flow near the secondary-secondary inlet **162** of the secondary-secondary heat exchanger **160**. Thus in an advantageous aspect, the secondary return conduit **144** may be configured to pass the secondary fluid sequentially over the primary-secondary heat exchange region **182** and the secondary-secondary heat exchange region **212** in counterflow to the supply direction in each case.

(24) As illustrated in FIG. 2, the primary-secondary heat exchanger **150** may be arranged sequentially with respect to the secondary-secondary heat exchanger **160**. Such a sequential arrangement may facilitate the low pressure stream of the secondary fluid to exchange heat with the primary fluid in the primary-secondary heat exchanger **150** prior to exchanging heat with the high pressure stream of the secondary fluid in the secondary-secondary heat exchanger **160**. In such a sequential arrangement, the primary-secondary inlet **152** may be positioned downstream (with respect to direction **166**) of the secondary-secondary outlet **164**. Further, the primary-secondary outlet **154** may also be positioned downstream (with respect to direction **166**) of the secondary-secondary outlet **164**.

(25) In certain aspects, the expanded low pressure stream of secondary fluid may first exchange heat with the high pressure stream of the primary fluid. For instance, in an embodiment (described further below), the secondary stream may flow over the primary-secondary heat exchanger **150** entirely before flowing over the secondary-secondary heat exchanger **160**. Thus the secondary return conduit may be configured to pass the expanded secondary fluid, sequentially over the entire primary-secondary heat exchange region (or each coil of the primary to secondary heat exchange coils before passing the secondary fluid over the secondary-secondary heat exchange coils.

(26) Referring back to FIGS. 1 and 2, in certain embodiments in which the cryoablation tool **100** includes a flexible catheter **102**, the secondary-secondary heat exchanger **160** and the primary-secondary heat exchanger **150** may each be positioned outside the catheter **102**, for instance, near the proximal section **106** of the cryoablation tool **100**. Further, the primary-primary heat exchanger **120** may be located within the flexible catheter **102**. This arrangement may further facilitate supplying the secondary fluid at pressures greater than the high pressure stream of the primary

fluid.

(27) As described previously, in embodiments where the cryoablation tool **100** includes a flexible catheter **102**, it may be advantageous to supply the primary fluid at pressures less than typical pressures for cryoablation. For instance, the primary fluid may be supplied at less than about 2000 psi, for instance, about 1800 psi, which, in the absence of the secondary fluid circuit **140**, may not result in sufficient cooling for cryoablation if the primary fluid is a cooling fluid. However, by passing the primary fluid through the primary-secondary heat exchanger **150**, recuperative heat exchange can be permitted between the secondary fluid to precool the primary fluid and achieve temperatures suitable for cryoablation even when the primary fluid is supplied at pressures less than the typical supply pressure for cryoablation.

(28) Advantageously, the high pressure stream of the secondary fluid may be at a pressure greater than a pressure of the high pressure stream of the primary fluid. The secondary fluid may, for instance, be supplied at pressures greater than about 2000 psi (e.g., 3500 psi). As the secondary fluid passes through the second cryocooler **170** (e.g., second J-T orifice **170**), the secondary fluid may reach cryogenic temperatures (e.g., less than about 190 Kelvin) if the secondary fluid is a cooling fluid. The expanded low pressure stream of the secondary fluid may remove heat from the high pressure stream of the primary fluid in the primary-secondary heat exchanger **150**, thereby precooling the primary fluid prior to the primary fluid's passage through the primary-primary heat exchanger **120**.

(29) FIG. **3** illustrates pictorial views of the primary-secondary and secondary-secondary heat exchangers according to an embodiment. FIG. **4** illustrates a pictorial view of the primary-primary heat exchanger **120** according to an embodiment.

(30) Referencing FIG. **3**, the primary supply conduit **112** may be a metal tube that may be fluidly coupled to the primary-secondary heat exchanger **150**. The primary-secondary heat exchanger **150** may include a plurality of first heat exchange coils **180** between the primary-secondary inlet **152** and the primary-secondary outlet **154**.

(31) Referring to FIGS. **2** and **3**, the plurality of first heat exchanger coils may define a first heat exchanger region **182**. The plurality of coils may include a first coil **184** and a second coil **186**. The first coil **184** may be upstream of the second coil **186** (with respect to direction **166**). The first coil **184** may be more proximal to the primary-secondary inlet **152** than to the primary-secondary outlet **154**. The second coil **186** may be more proximal to the primary-secondary outlet **154** than to the primary-secondary inlet **152**. The primary-secondary heat exchange region may extend between the first coil **184** and the second coil **186** and/or between the primary-secondary inlet **152** and the primary-secondary outlet **154**.

(32) According to certain embodiments, the plurality of first heat exchanger coils **180** (primary-secondary heat exchanger) may be formed from a metal tube (a tubular region of the primary supply conduit) wound about a mandrel **188**. The metal tube may include an outer wall. The outer wall may include extensions to increase a surface area of the outer wall. For instance, in an embodiment, the first heat exchange coils **180** may include fins **190** to increase heat exchange over the first heat exchanger region **182**. The primary fluid may flow through an interior of the metal tube. Referring to FIGS. **2** and **3**, the expanded low pressure stream of primary fluid may flow over the finned portions of the metal tube. The fins **190** may enhance heat exchange between the high pressure stream of the primary fluid flowing through the interior of the metal tube, and the low pressure stream of the secondary fluid flowing over the exterior of the finned portions of the metal tube.

(33) Referencing FIG. **2**, the primary fluid, after exchanging heat with the secondary fluid in the primary-secondary exchanger, may continue toward the primary-primary heat exchanger **120**. Referring to FIG. **4**, the primary-primary heat exchanger **120** may include a plurality of second heat exchange coils **200**. With reference to FIGS. **2** and **4**, the primary-secondary heat exchanger **150** and the primary-primary heat exchanger **120** may be in a sequential arrangement. Thus, each coil

of the plurality of second heat exchange coils **200** may be downstream (with respect to direction **166**) of the first coil **184** of the plurality of first heat exchange coils **180**.

(34) According to certain embodiments, the plurality of second heat exchange coils **200** (primary-primary heat exchanger) may be formed from a metal tube (tubular region of the primary supply conduit) wound about a mandrel (e.g., analogous to the mandrel **188** of FIG. **3**). Returning to FIG. **4**, the metal tube may include an outer wall. The outer wall may include extensions to increase a surface area of the outer wall. For instance, in an embodiment, the second heat exchange coils **200** may include fins **202** to increase heat exchange over the second heat exchanger region **204**. The primary fluid may flow through an interior of the metal tube. Referring to FIGS. **2** and **3**, the expanded low pressure stream of primary fluid may flow over the finned portions of the metal tube. The fins **202** may enhance heat exchange between the high pressure stream of the primary fluid flowing through the interior of the metal tube, and the low pressure stream of the primary fluid flowing over the exterior of the finned portions of the metal tube.

(35) The primary fluid leaving the primary-primary heat exchanger **120** may enter the primary conduit **128** (e.g., primary capillary tube), and flow toward the first cryocooler **130**, as described previously. The first cryocooler **130** may, in some instances, be a first J-T orifice **130**. The primary fluid may expand in the primary expansion chamber positioned downstream of the primary-primary heat exchange region near the distal operating tip **108** for expansion of the primary fluid. The expanded low pressure primary fluid may be carried by the primary return conduit **114**.

(36) Referring again to FIGS. **2** and **4**, the second heat exchange region (the primary-primary heat exchange region) **204** may be arranged within the primary return conduit **114**. Thus the primary return conduit **114** may generally surround the plurality of second heat exchange coils (second heat exchange region) **200**. The primary return conduit **114** may thus pass the expanded low pressure primary fluid over the second heat exchange region and thereby exchange heat (e.g., recuperative heat exchange) with the high pressure stream of primary fluid flowing in the interior of the metal tube of the plurality of second heat exchange coils **200** (primary-primary heat exchanger).

(37) Advantageously, as illustrated in FIG. **2**, the primary return conduit **114** is configured to pass the primary fluid over the primary-primary heat exchange region **204** but not the primary-secondary heat exchange region **182** or the secondary-secondary heat exchange region **212**. Advantageously, the primary-secondary heat exchange region **182** and the secondary-secondary heat exchange region **212** are each disposed within the secondary return conduit **144**, the secondary return conduit fluidly isolating the expanded primary fluid flowing in the primary return conduit **114** from the primary-secondary heat exchange region **182** and the secondary-secondary heat exchange region **212**. Advantageously the primary return conduit **114** may comprise a portion disposed concentrically about the secondary return conduit **144**. This enables the flow of primary fluid to pass around the secondary return conduit **144** without passing over either of the primary secondary **182** or the secondary-secondary **212** heat exchange regions disposed within the secondary return conduit **144**. Advantageously this portion is downstream of the primary-primary heat exchange region **204**, with respect to primary fluid flow.

(38) With reference to FIGS. **2** and **3**, the secondary supply conduit **142** may supply the secondary fluid to the secondary-secondary heat exchanger **160**. The secondary-secondary heat exchanger **160** may include a plurality of third heat exchange coils **210** that may define a secondary-secondary heat exchange region **212**.

(39) According to certain illustrated embodiments, the plurality of third heat exchanger coils (secondary-secondary heat exchanger) may be formed from a metal tube (tubular portion of the second supply conduit) wound about a mandrel. In some cases, the mandrel about which the third heat exchanger coils may be wound may either be identical to, or in the alternative, coextensive with the mandrel **188** about which the first heat exchanger coils may be wound. Alternatively, the mandrel about which third heat exchanger coils may be wound may be separate from the mandrel **188** of about which the first heat exchanger coils may be wound.

(40) Returning to FIG. 3, the metal tube of the third heat exchange coils **210** may include an outer wall. The outer wall may include extensions to increase a surface area of the outer wall. For instance, in an embodiment, the third heat exchange coils **210** may include fins **214** to increase heat exchange over the third heat exchanger region. The secondary fluid entering the secondary-secondary inlet **162** may flow through an interior of the metal tube. Referring to FIGS. 2 and 3, the expanded low pressure stream of secondary fluid (leaving the second J-T orifice **170**) may flow over the finned portions of the metal tube. The fins **214** may enhance heat exchange between the high pressure stream of the secondary fluid flowing through the interior of the metal tube, and the low pressure stream of the secondary fluid flowing over the exterior of the finned portions of the metal tube.

(41) As described previously, the low pressure stream of the secondary fluid may first exchange heat with the high pressure stream of the primary fluid in the primary-secondary heat exchanger **150**, prior to exchanging heat with the high pressure stream of the secondary fluid in the secondary-secondary heat exchanger **160**. Accordingly, the each coil of the plurality of third heat exchange coils **210** may be positioned upstream of the first coil **184** of the plurality of first heat exchange coils **180**. Thus the secondary-secondary heat exchange region **212** and primary-secondary heat exchange region **182** may be arranged within the secondary return conduit **144** and the secondary-secondary heat exchange region **212** may be arranged downstream to the primary-secondary heat exchange region **182**, with respect to the flow direction within the secondary return conduit **144**.

(42) The secondary fluid leaving the secondary-secondary heat exchanger **160** may enter the secondary conduit **168** (e.g., secondary capillary tube **168**), and flow toward the second cryocooler **170**, as described previously. The second cryocooler **170** may, in some instances, be the second J-T orifice **170**. The secondary fluid may expand in the secondary expansion chamber positioned downstream of the primary-secondary heat exchange region. Upon expansion, the low pressure stream of the secondary fluid flows over the finned portion of the primary-secondary heat exchange region for recuperative heat exchange with the high pressure stream of the primary fluid. The low pressure stream of the secondary fluid then flows over the finned portion of the secondary-secondary heat exchange region **212** for recuperative heat exchange with the high pressure stream of the secondary fluid.

(43) Advantageously, in the instances where the primary fluid and the secondary fluid are each cooling fluids, the secondary fluid may reach cryogenic temperatures after flowing through the second cryocooler **170**. As the secondary fluid flows over the finned tube of the first heat exchange coils **180**, the temperature of the secondary fluid may gradually increase. The secondary fluid flowing over the finned tube of the third heat exchange coils **210** (e.g., near the secondary-secondary outlet **164**) may thus be at a higher temperature than the temperature of the secondary fluid flowing just proximal to the primary-secondary outlet **154**. Thus, by passing the secondary fluid first over the primary-secondary heat exchanger **150**, the secondary fluid at its coldest temperature may be in heat exchange with the primary fluid, thereby permitting optimal heat removal from the primary fluid for effective precooling thereof.

(44) Referring again to FIG. 2, the secondary return conduit **144** may generally surround the plurality of third heat exchange coils **210**. Advantageously, the secondary return conduit **144** may be fluidly isolated from the primary return conduit **114**, so as to fluidly isolate the low pressure stream of the primary fluid from the low pressure stream of the secondary fluid. Thus, the low pressure stream of the primary fluid may not flow over the secondary-secondary heat exchanger **160** or over the primary-secondary heat exchanger **150**.

(45) According to advantageous embodiments, the primary-secondary heat exchange region, the primary-primary heat exchange region and the secondary-secondary heat exchange region **212** may be longitudinally (e.g., along direction **166**) separated from each other. Referencing FIGS. 2-4, each coil of the secondary-secondary heat exchange region **212** may be upstream of the primary-secondary heat exchange region and the primary-primary heat exchange region. Further, the

distance between adjacent coils of the plurality of third heat exchange coils **210** may be substantially less than the distance between a coil of the secondary-secondary heat exchange region **212** and a coil of the primary-secondary heat exchange region or a coil of the primary-primary heat exchange region.

(46) Continuing with FIGS. **2-4**, each coil of the plurality of third heat exchange coils **210** (forming the secondary-secondary heat exchanger **160**) may be upstream (with respect to direction **166**) of each of the primary-secondary inlet **152** and the primary-primary inlet **122**. Further, each coil of the plurality of first coils (forming the primary-secondary heat exchanger **150**) may be upstream (with respect to direction **166**) of the primary-primary inlet **122**.

(47) Arrangements such as those disclosed above may facilitate effective recuperative heat exchange, thereby permitting desired cryoablation (or thawing) properties at the distal operating tip **108**, as described previously.

Example

(48) FIGS. **5** and **6** illustrate a thermodynamic schematic and a thermodynamic property diagram to illustrate an example of the dual stage cryocooler. According to the example, the primary fluid can be argon. The secondary fluid can also be argon. Referencing FIG. **5**, the primary fluid circuit **110** is illustrated by solid lines, while the secondary fluid circuit **140** is illustrated by dotted lines.

(49) Referencing FIGS. **5** and **6**, the secondary fluid is at a thermodynamic state **1** at the secondary-secondary inlet **162** as the secondary fluid enters the secondary-secondary heat exchanger **160**. The secondary fluid is at thermodynamic state **2** at the secondary-secondary outlet **164**. The secondary fluid undergoes expansion and reaches thermodynamic state **3**. The secondary fluid at thermodynamic state **3** exchanges heat over the primary-secondary heat exchanger **150**. The secondary fluid is at thermodynamic state **4** after heat exchange over the primary-secondary heat exchanger **150**. The secondary fluid at thermodynamic state **4** flows over the secondary-secondary heat exchanger **160**, and as the secondary fluid flows past the secondary-secondary heat exchanger **160**, reaches thermodynamic state **5**, after exchanging heat with the secondary fluid entering the secondary-secondary heat exchanger **160** at thermodynamic state **1**.

(50) Referencing FIGS. **5** and **6**, the primary fluid is at a thermodynamic state **6** at the primary-secondary inlet **152** as the primary fluid enters the primary-secondary heat exchanger **150**. The primary fluid is at thermodynamic state **7** at the primary-secondary outlet **154**. The primary fluid enters the primary-primary inlet **122** at thermodynamic state **7**, and leaves the primary-primary outlet **124** at thermodynamic state **8**. The primary fluid undergoes expansion and reaches thermodynamic state **9**. The primary fluid at thermodynamic state **9** exchanges heat with tissue. The primary fluid reaches thermodynamic state **10** after heat exchange with the tissue. The primary fluid at thermodynamic state **10** flows over the primary-primary heat exchanger **120**, and reaches thermodynamic state **11** after exchanging heat with the primary fluid entering the secondary-secondary heat exchanger **160** at thermodynamic state **7**.

(51) In the example of FIGS. **5** and **6**, the primary fluid and the secondary fluid may each be a cooling fluid. Thus, expansion of the primary fluid between states **8** and **9**, and the expansion of the secondary fluid between states **2** and **3** each result in cooling. As seen from FIG. **6**, the primary fluid entering the primary-secondary heat exchanger **150** at thermodynamic state **6** may be at a lower pressure than the secondary fluid entering the secondary-secondary heat exchanger **160** at thermodynamic state **1**. However, the temperature of the primary fluid after expansion, at thermodynamic state **9** may be less than the temperature of the secondary fluid after expansion, at thermodynamic state **3**. Thus, despite supplying the primary fluid at lower pressure, as a result of dual-stage cryocooling, low temperatures suitable for cryoablation (e.g., less than 190 K, for instance, about 120 K) may be achieved.

Claims

1. A cryoablation tool comprising: a shaft having a proximal end and a distal end; a primary supply conduit configured to supply a primary fluid from a high-pressure cryogenic gas source to a distal end of the shaft, the primary supply conduit comprising a first Joule-Thomson (J-T) orifice, configured to deliver the primary fluid to a first expansion chamber; a primary return conduit configured to carry the primary fluid away from the first expansion chamber; a secondary supply conduit configured to supply a secondary fluid to a second J-T orifice, the second J-T orifice configured to deliver the secondary fluid to a second expansion chamber; and a secondary return conduit configured to carry the secondary fluid away from the second expansion chamber, the primary supply conduit comprising a primary-secondary heat exchange region and a primary-primary heat exchange region, the primary-primary heat exchange region being down stream of the primary-secondary heat exchange region; and the secondary supply conduit comprising a secondary-secondary heat exchange region up stream of the second J-T orifice, and the primary return conduit further configured to pass the primary fluid over the primary-primary heat exchange region, in counterflow to the supply direction; the secondary return conduit further configured to pass the secondary fluid sequentially over the primary-secondary heat exchange region and the secondary-secondary heat exchange region in counterflow to the supply direction.
2. The cryoablation tool according to claim 1, wherein the secondary return conduit is configured to pass the secondary fluid sequentially over a full length of the primary-secondary heat exchange region before passing the secondary fluid over a full length of the secondary-secondary heat exchange region.
3. The cryoablation tool according to claim 1, wherein the secondary-secondary heat exchange region and primary-secondary heat exchange region are arranged within the secondary return conduit and the secondary-secondary heat exchange region is arranged downstream to the primary-secondary heat exchange region with respect to a flow direction within the secondary return conduit.
4. The cryoablation tool according to claim 1, wherein the primary supply conduit comprises a tubular region wound in a series of turns about a mandrel to form the primary-secondary heat exchanger, and the secondary supply conduit comprises a tubular region wound in a series of turns about the same mandrel to form the secondary-secondary heat exchanger.
5. The cryoablation tool according to claim 1, wherein the primary return conduit is configured to pass the primary fluid over the primary-primary heat exchange region but not the primary-secondary heat exchange region or the secondary-secondary heat exchange region.
6. The cryoablation tool according to claim 1, the primary return conduit is configured to pass the primary fluid over the primary-primary heat exchanger; and the secondary return conduit is configured to pass the secondary fluid over the primary-secondary heat exchanger prior to passing the secondary fluid over the secondary-secondary heat exchanger.
7. The cryoablation tool according to claim 1, wherein the primary supply conduit is arranged in a first plurality of coils, configured as a primary-secondary heat exchanger within the primary-secondary heat exchange region and a second plurality of coils, configured as a primary-primary heat exchanger, within the primary-primary heat exchange region; and the secondary supply conduit being arranged in a third plurality of coils configured as a secondary-secondary heat exchanger within the secondary-secondary heat exchange region.
8. The cryoablation tool according to claim 7, wherein the first plurality of coils comprises a first coil and a second coil, the first coil arranged upstream of the second coil relative to the direction of primary fluid flow through the first coil; each coil of the second plurality of coils being positioned downstream of the said second coil relative to the direction of primary fluid flow in the first coil; and each coil of the third plurality of coils is positioned upstream of the first coil relative to the direction of primary fluid flow in the said first coil.
9. The cryoablation tool according to claim 1, wherein the secondary return conduit is configured

to pass the secondary fluid sequentially over a full length of the primary-secondary heat exchange region before passing the secondary fluid over a full length of the secondary-secondary heat exchange region, the secondary return conduit is configured to pass the secondary fluid over each coil of the primary-secondary heat exchanger before passing it over the secondary to secondary heat exchanger.

10. The cryoablation tool according to claim 1, wherein the primary-primary heat exchange region is arranged within the primary return conduit and is further, arranged distally of both the primary-secondary and the secondary-secondary heat exchange regions.

11. The cryoablation tool according to claim 1, wherein the secondary-secondary heat exchange region is arranged proximally to the primary-secondary heat exchange region, within the secondary return conduit.

12. The cryoablation tool according to claim 1, wherein the primary supply conduit comprises a tubular region wound in a series of turns about a mandrel to form the primary-secondary heat exchanger, the secondary supply conduit may comprise a tubular region wound in a series of turns about a mandrel to form the secondary-secondary heat exchanger, the primary supply conduit may comprise a tubular region wound in a series of turns about a mandrel to form the primary-secondary heat exchanger, and the secondary supply conduit may comprise a tubular region wound in a series of turns about the same mandrel to form the secondary-secondary heat exchanger; the primary supply conduit may comprise a tubular region wound in a series of turns about a mandrel to form the primary-primary heat exchanger.

13. The cryoablation tool according to claim 1, wherein the primary-secondary heat exchange region and the secondary-secondary heat exchange region are each disposed within the secondary return conduit; the secondary return conduit may then fluidly isolate the primary return conduit from the primary-secondary heat exchange region and the secondary-secondary heat exchange region.

14. The cryoablation tool according to claim 1, wherein the primary return conduit comprises a portion disposed concentrically about the secondary return conduit, the portion is downstream of the primary-primary heat exchange region with respect to the direction of expanded primary fluid flow.

15. The cryoablation tool according to claim 1, wherein the primary return conduit is configured to vent the primary fluid to atmosphere and/or the secondary return conduit is configured to vent the secondary fluid to atmosphere.

16. A cryoablation method using a cryoablation tool with a shaft having arranged therein a primary supply conduit connected to a first Joule-Thomson (J-T) orifice in a first expansion chamber and a secondary supply conduit connected to a second J-T orifice in a second expansion chamber, the method comprising: supplying a primary fluid from a high-pressure cryogenic gas source to a distal end of the shaft; expanding, cryogenically, a high-pressure stream of a primary fluid to a low-pressure stream of the primary fluid via the primary fluid circuit, the first J-T orifice being fluidly coupled to a primary-primary heat exchanger via a primary supply conduit to receive the high-pressure stream of the primary fluid after the high-pressure stream of the primary fluid passes through the primary-primary heat exchanger; directing the primary fluid away from the first expansion chamber via a primary return conduit arranged within the shaft; directing a low-pressure stream of a secondary fluid to exchange heat with a high-pressure stream of a primary fluid in a primary-secondary heat exchanger prior to exchanging heat with the high-pressure stream of the secondary fluid in a secondary-secondary heat exchanger; directing the secondary fluid leaving the secondary-secondary heat exchanger to enter a secondary conduit and to flow toward the second J-T orifice; and directing the secondary fluid away from the second expansion chamber via a secondary return conduit arranged within the shaft.

17. The method of claim 16, further comprising directing, via the secondary return conduit, the secondary fluid sequentially over the primary-secondary heat exchanger and the secondary-

secondary heat exchanger in counterflow to the supply direction in each case.

18. The method of claim 16, further comprising upon expansion, directing the low-pressure stream of the secondary fluid to flow over finned portions of the primary-secondary heat exchanger for recuperative heat exchange with the high-pressure stream of the primary fluid.

19. The method of claim 18, further comprising directing the low-pressure stream of the secondary fluid to flow over a finned portion of the secondary-secondary heat exchanger for recuperative heat exchange with the high-pressure stream of the secondary fluid.

20. The method of claim 16, wherein the secondary fluid is prioritized to flow first through the primary-secondary heat exchanger such that the secondary fluid is coldest when in heat exchange with the primary fluid, thereby permitting optimal heat removal from the primary fluid for effective precooling of the primary fluid.
